



INTRODUCTION TO MCP SERVERS

SID SINGH, 19.05.2026





I'm going to a tech conference next month — which talks overlap most with what the group is interested in?



*I'm going to **a tech conference** next month — which talks overlap most with what the group is interested in?*



*I'm going to a tech conference **next month** — which talks overlap most with what the group is interested in?*



*I'm going to a tech conference next month — **which talks overlap most** with what the group is interested in?*



*I'm going to a tech conference next month — which talks overlap most with **what the group is interested in?***

GOALS

- By the end of the workshop, your LLM should be able to answer this question.
- Demonstrate the ecosystem around MCP servers.
- You should understand how you can implement agents (using MCP) for your own use cases.

MCP: MODEL CONTEXT PROTOCOL

- These are protocols that can be used to communicate to LLMs in a standardized way.
- MCP servers ensure that models have access to relevant context to improve predictions, reasoning, and continuity in workflows.

MCP: CORE CONCEPTS

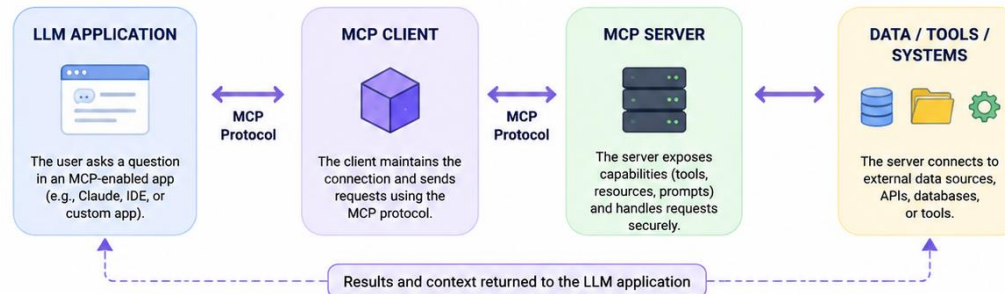
- Model Management
 - Hosts multiple AI/ML models centrally.
 - Keeps track of version, configurations, and metadata
- Context Handling
 - Maintains the state or context for each session or request
 - Ensures models can make context-aware decisions
- Interoperability
 - Exposes APIs for applications to query models
 - Standardizes how models receive and return context across systems
- Scalability
 - Can handle many concurrent users or applications accessing models
 - Optimizes resource allocation for CPU/GPU usage

MCP SERVERS

(MODEL CONTEXT PROTOCOL)

MCP is an **open protocol** that standardizes how apps provide context to LLMs.
MCP Servers act as **secure bridges** between LLMs and external data, tools, and systems.

HOW MCP WORKS



WHAT CAN MCP SERVERS DO?

- Expose Tools**
Allow the LLM to take actions (e.g., search, run a query, send an email).
- Provide Resources**
Give the LLM access to data (e.g., files, docs, DB records).
- Offer Prompts**
Provide reusable prompt templates and instructions.
- Ensure Security**
Control access, permissions, and data boundaries.

EXAMPLES OF MCP SERVERS

- Google Drive MCP Server**
Search and access files.
- Slack MCP Server**
Read messages, channels, and send updates.
- PostgreSQL MCP Server**
Query your databases safely.
- GitHub MCP Server**
Search repos, issues, and pull requests.

WHY MCP?

- Standardization**
One protocol, many integrations.
- Security & Control**
You decide what data and tools are accessible.
- Flexibility**
Connect any data source or system.
- Ecosystem Growth**
A growing community of servers and clients.

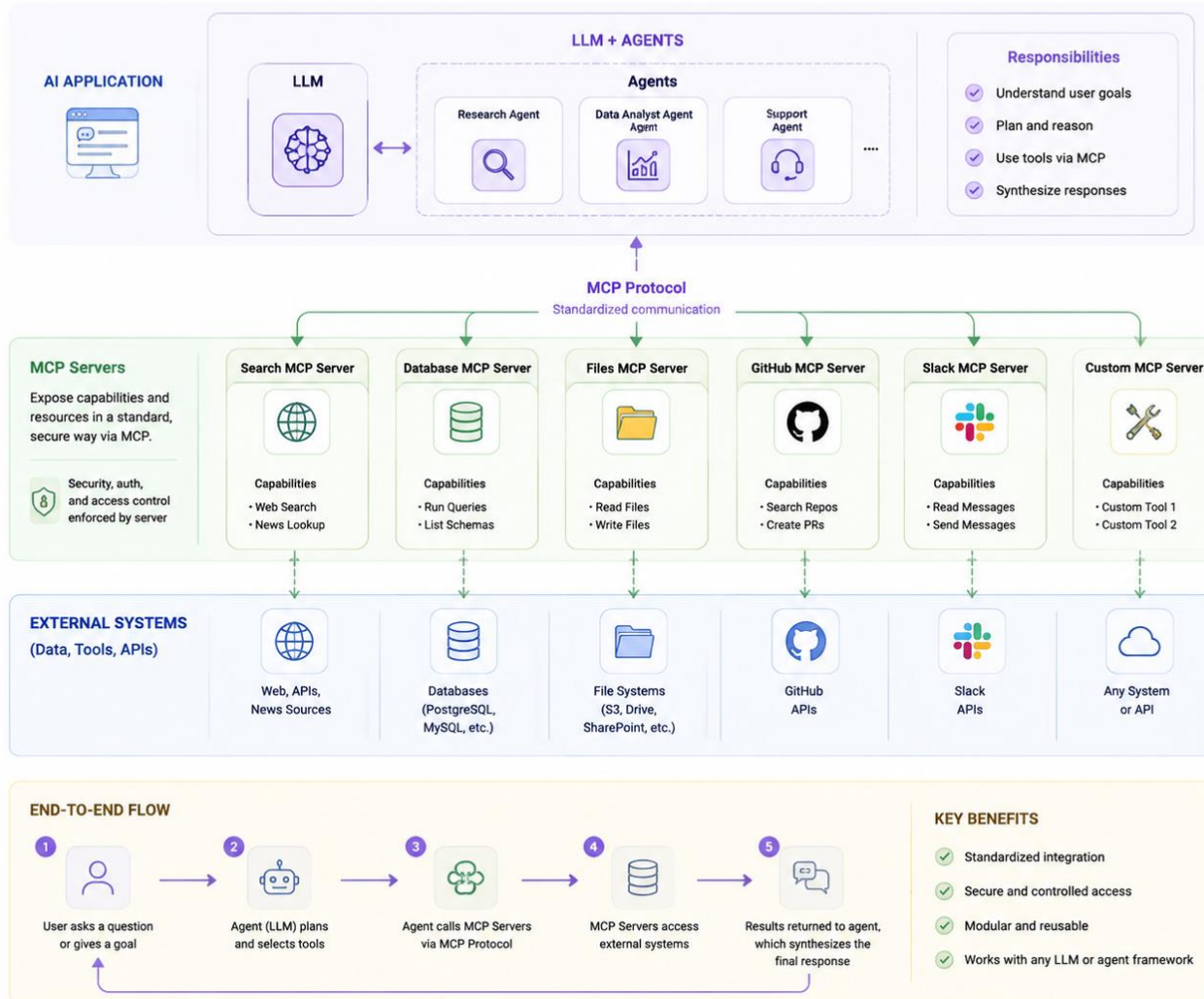
KEY TAKEAWAYS

- MCP is the USB-C for AI applications.
- MCP Servers bridge LLMs with real-world context and tools.
- It's open, extensible, and community-driven.
- MCP unlocks more powerful, useful, and context-aware AI apps.

OVERVIEW

MCP Architecture: How LLMs, Agents and MCP Servers Work Together

MCP (Model Context Protocol) standardizes how AI applications connect to external data, tools and systems through MCP Servers.

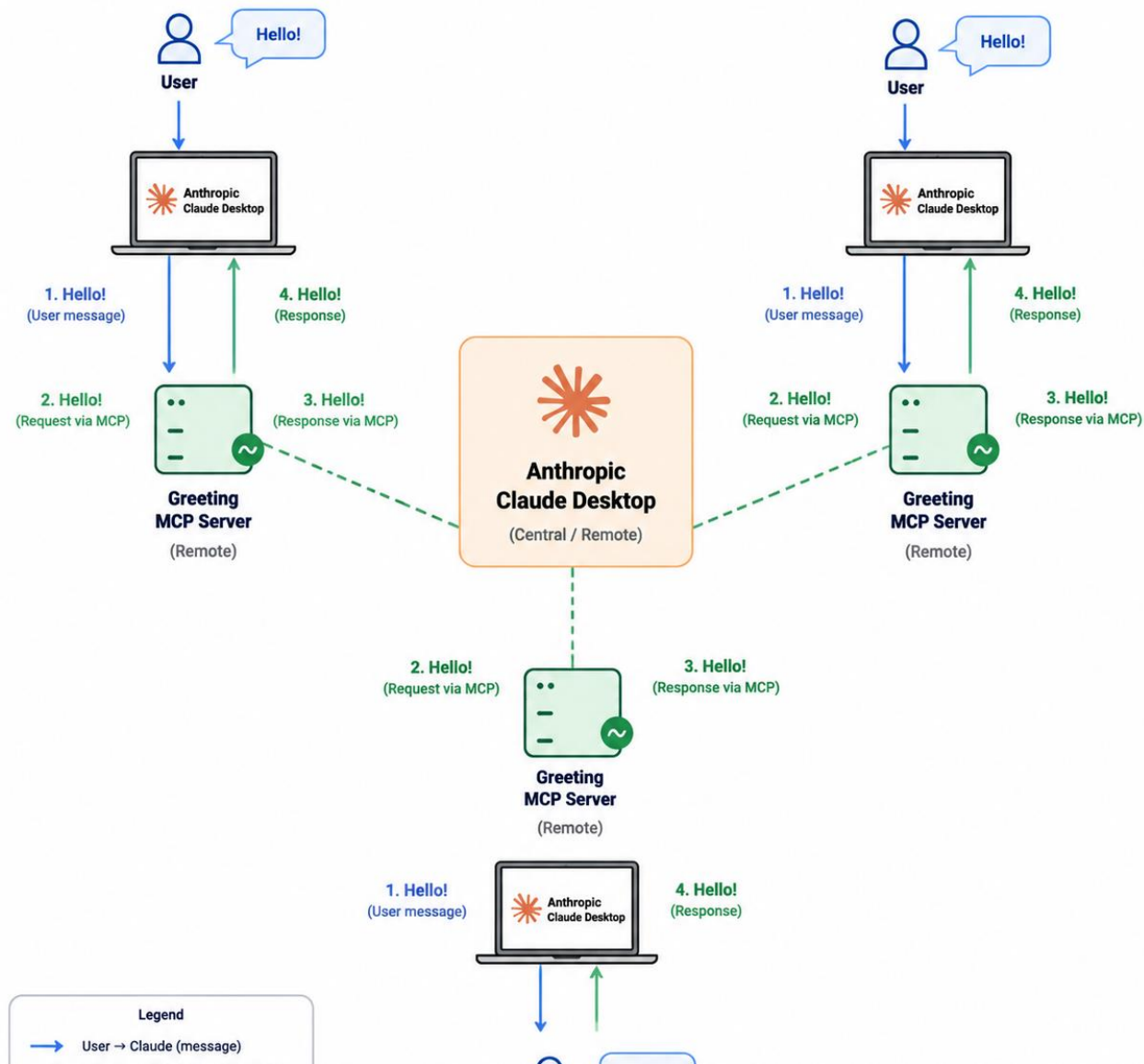


MCP SERVERS IN AGENTIC SYSTEMS

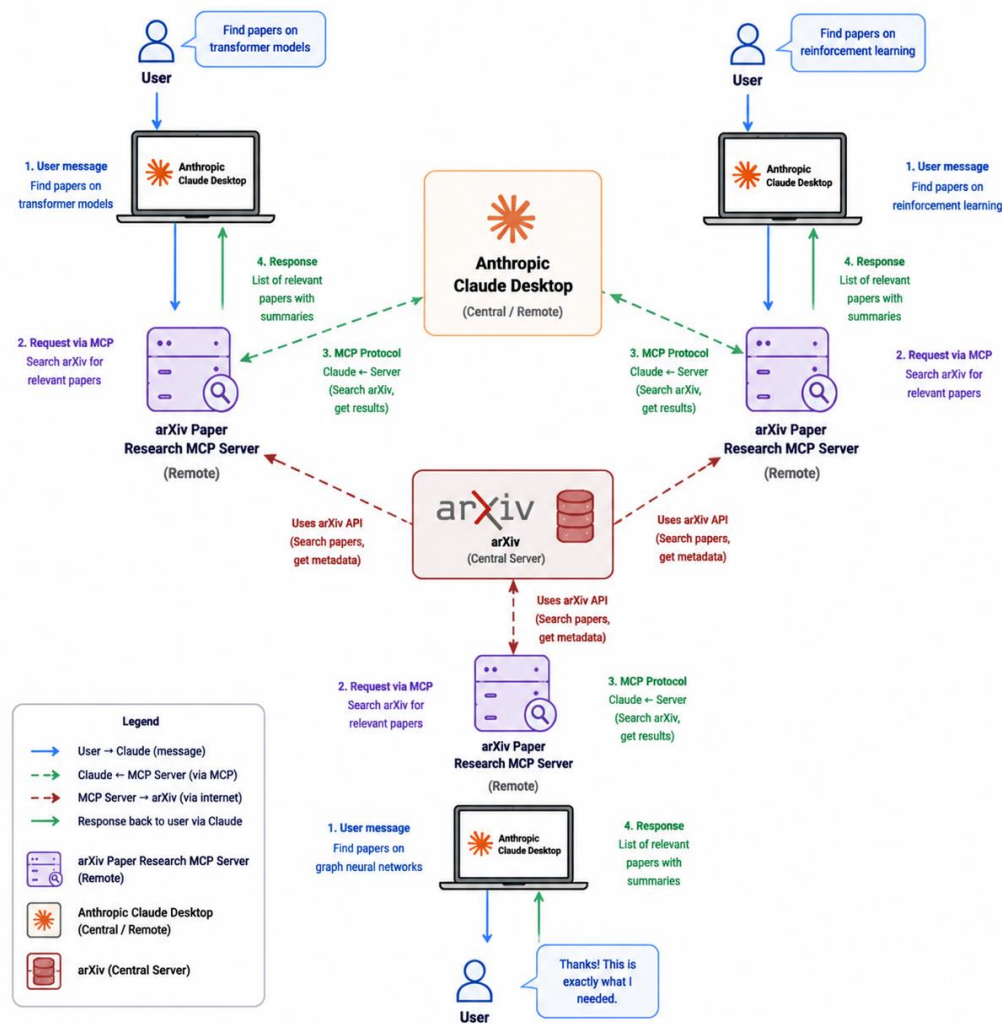
METHOD

- 6 levels of difficulty

Level	New concept	Data source	Magic demonstrated
1	Tool declaration & discovery	Hardcoded list	Claude can call tools
2	Live external data	arXiv API	Answers change in real time
3	Memory across sessions	Local SQLite	Claude remembers you
4	Shared remote service	Deployed server	Multiple users, live data
5	Tool design as UX	confs.tech JSON	Design quality is measurable
6	Multi-server synthesis	All three combined	Claude acts like an assistant



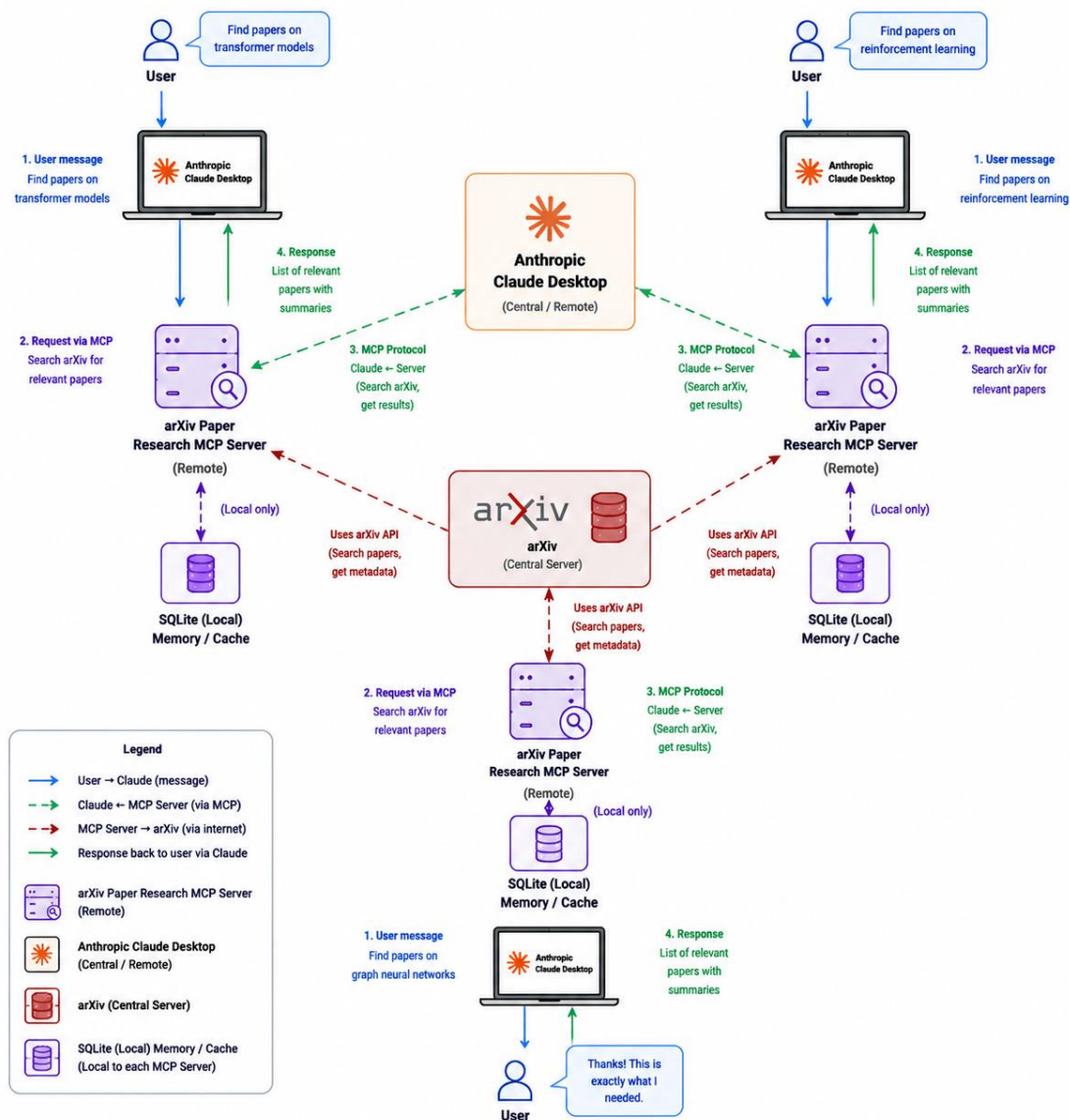
ARCHITECTURE: LEVEL I



ARCHITECTURE LEVEL II

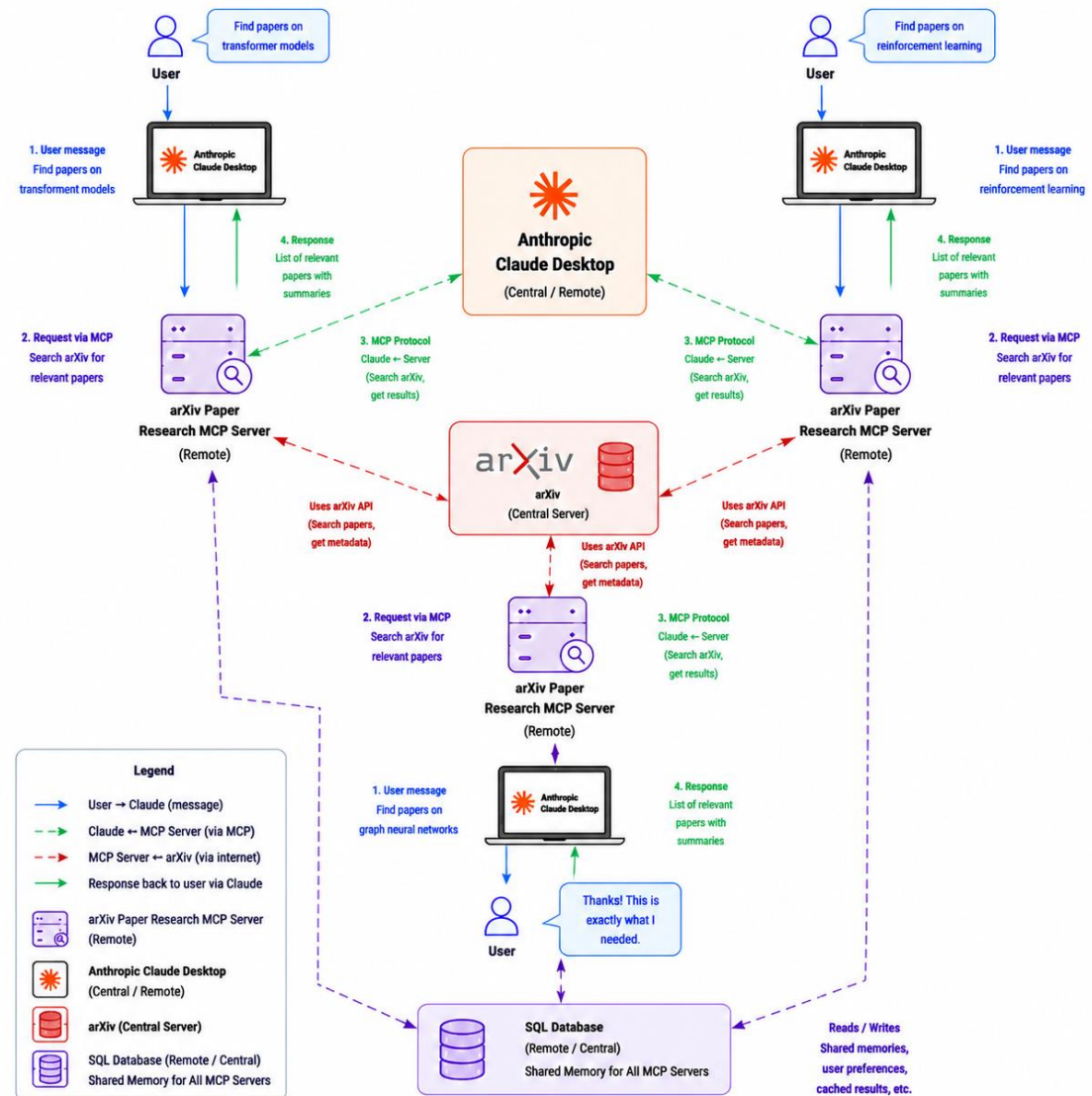
FRICTION: BUILT-IN SEARCH VS. EXTERNAL TOOLS

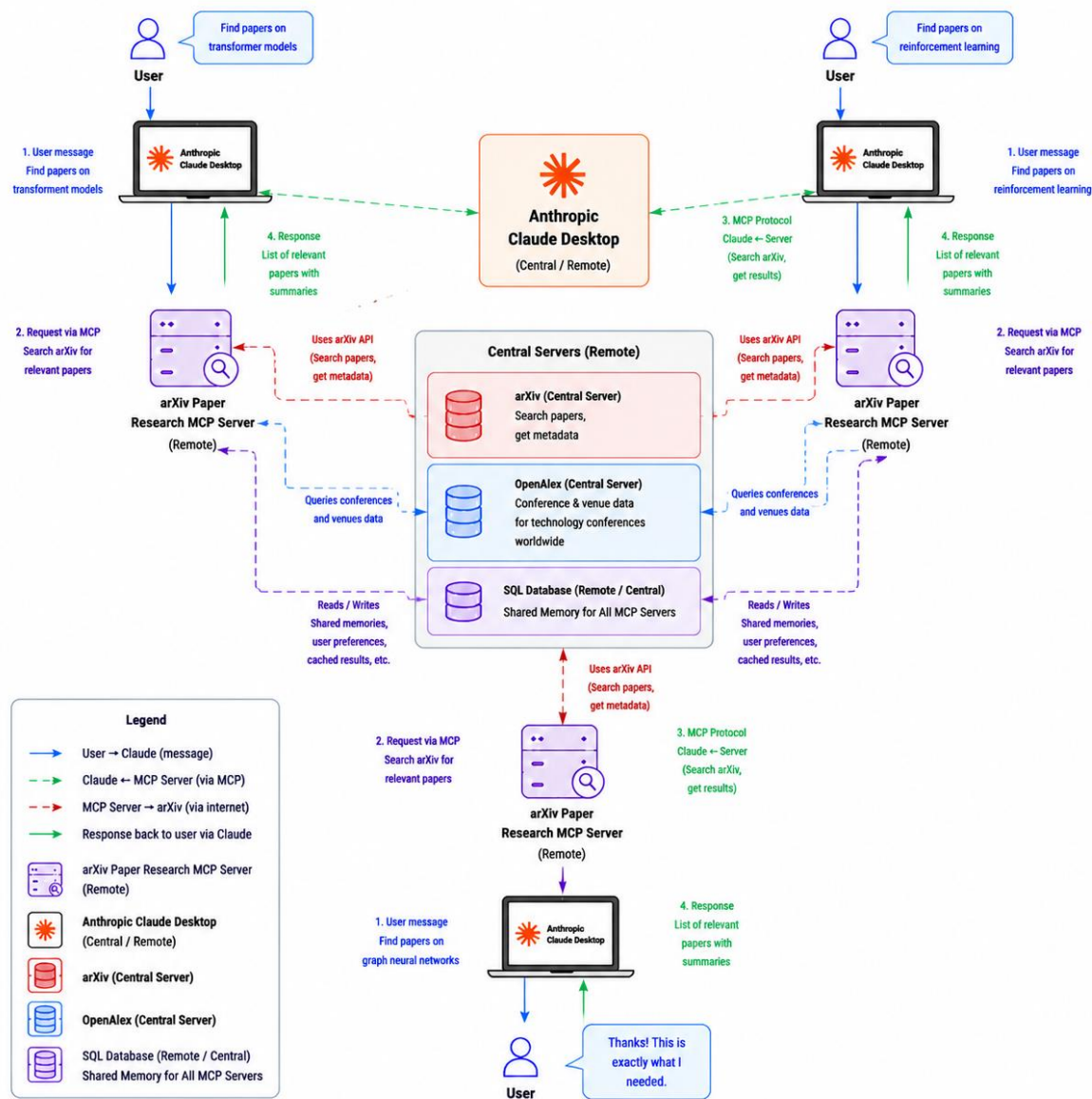
- In Level2 complexity, Claude fails often to use the tool you designed, Why?
- There is a built-in search that is aggressively prioritized. The failure demonstrates that your design is not good.
- Tool design is not just an engineering problem!



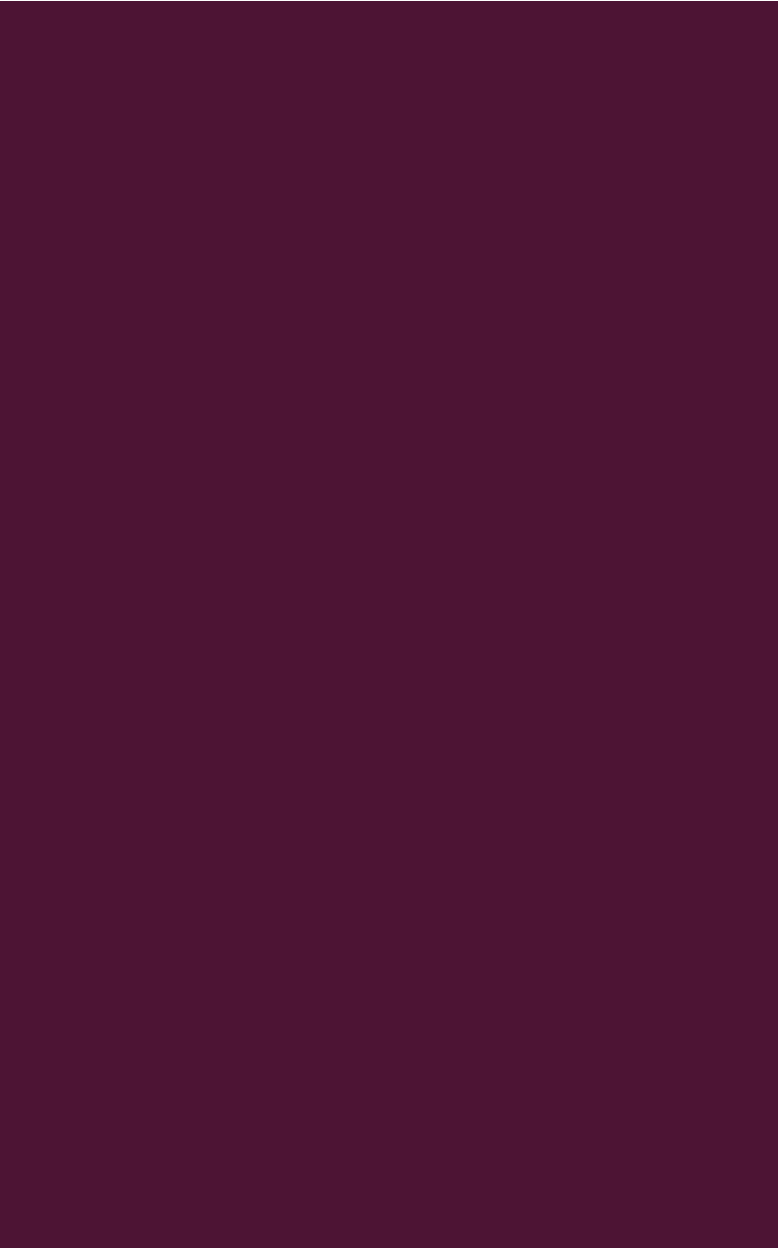
ARCHITECTURE LEVEL III

ARCHITECTURE LEVEL IV





ARCHITECTURE LEVEL V



END OF LECTURE